



Introduction

Department of Computer Engineering

Sharif University of Technology

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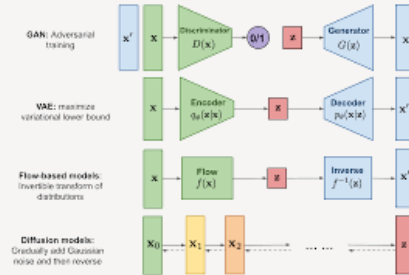
Maryam Ramezani maryam.ramezani@sharif.edu



Course Description

How is our course?

Machine Learning Mathematics





Machine Learning Mathematics

01

Linear Algebra

Provides the foundation for manipulating data in high-dimensional spaces, essential for vector operations in machine learning.

02

Statistics & Probability

Offers tools to model uncertainty and make inferences about data, forming the backbone of many machine learning algorithms.

03

Optimization

Focuses on finding the best parameters for a model by minimizing or maximizing an objective function.

04



Information Theory

Analyzes data compression and transmission, helping quantify uncertainty and information gain in machine learning models.

05

Signal Processing

Involves analyzing and transforming data signals, important for handling time-series data and feature extraction in machine learning.

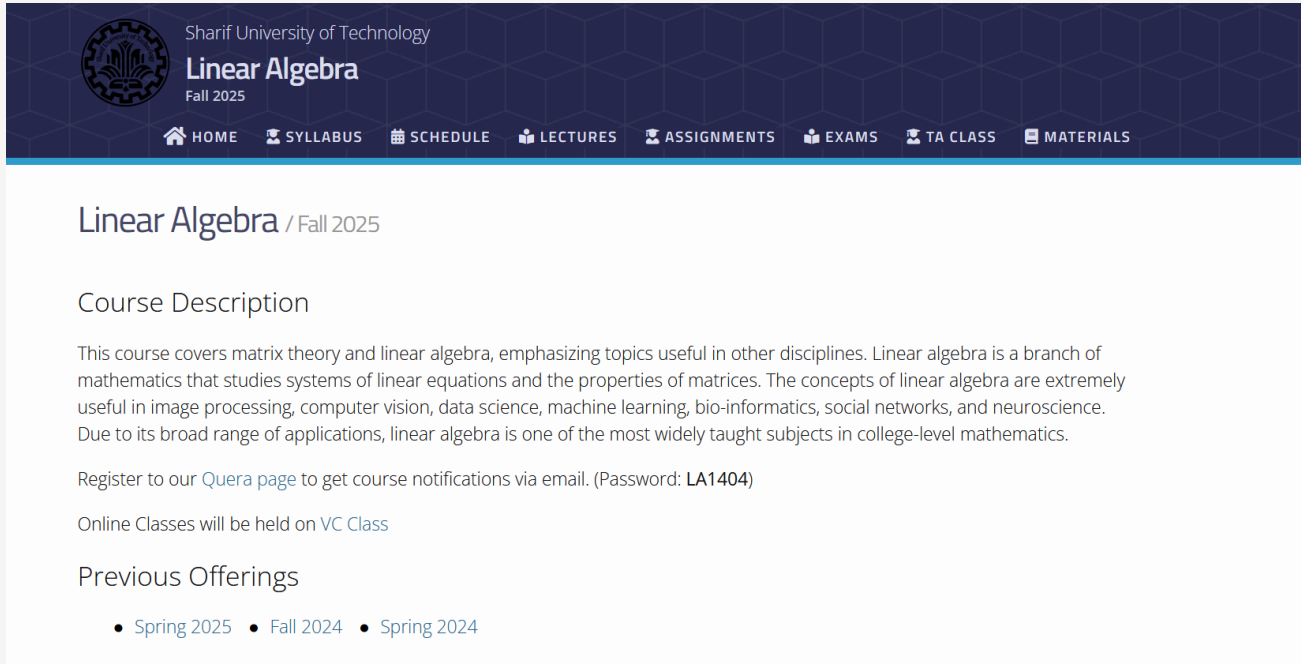
06

Graph Theory

Studies networks of nodes and edges, crucial for understanding relationships in data and complex models like Graph Neural Networks.

Course Material

<https://sut-ce-courses.github.io/linearalgebra> (jabrekh.ir)



The screenshot shows the homepage of the Linear Algebra course website. The header is dark blue with the Sharif University of Technology logo and the course title 'Linear Algebra Fall 2025'. A navigation bar contains links for HOME, SYLLABUS, SCHEDULE, LECTURES, ASSIGNMENTS, EXAMS, TA CLASS, and MATERIALS. The main content area has a white background and includes the course title 'Linear Algebra / Fall 2025', a 'Course Description' section with a paragraph about the course's focus on matrix theory and its applications, a registration link 'Quera page' with password 'LA1404', a note about online classes on 'VC Class', and a 'Previous Offerings' section with links for Spring 2025, Fall 2024, and Spring 2024.

Sharif University of Technology

Linear Algebra

Fall 2025

[HOME](#) [SYLLABUS](#) [SCHEDULE](#) [LECTURES](#) [ASSIGNMENTS](#) [EXAMS](#) [TA CLASS](#) [MATERIALS](#)

Linear Algebra / Fall 2025

Course Description

This course covers matrix theory and linear algebra, emphasizing topics useful in other disciplines. Linear algebra is a branch of mathematics that studies systems of linear equations and the properties of matrices. The concepts of linear algebra are extremely useful in image processing, computer vision, data science, machine learning, bio-informatics, social networks, and neuroscience. Due to its broad range of applications, linear algebra is one of the most widely taught subjects in college-level mathematics.

Register to our [Quera page](#) to get course notifications via email. (Password: **LA1404**)

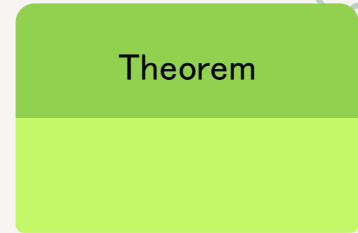
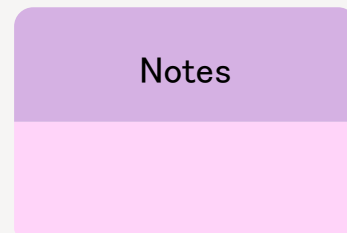
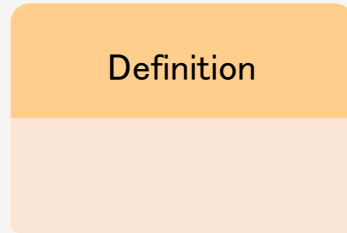
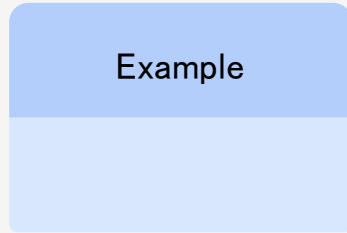
Online Classes will be held on [VC Class](#)

Previous Offerings

- [Spring 2025](#)
- [Fall 2024](#)
- [Spring 2024](#)

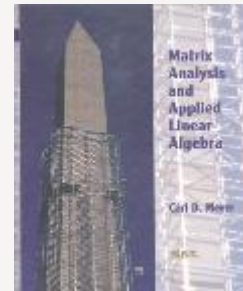
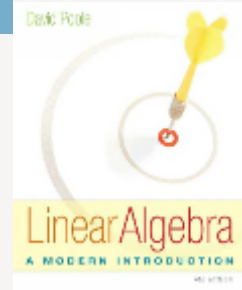
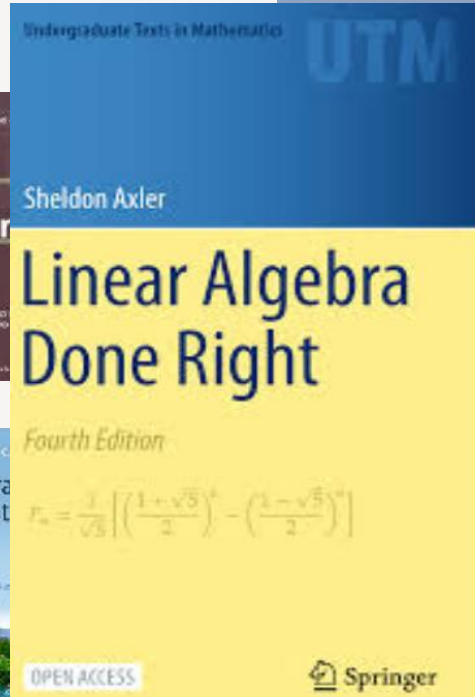
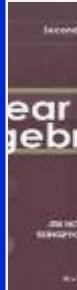
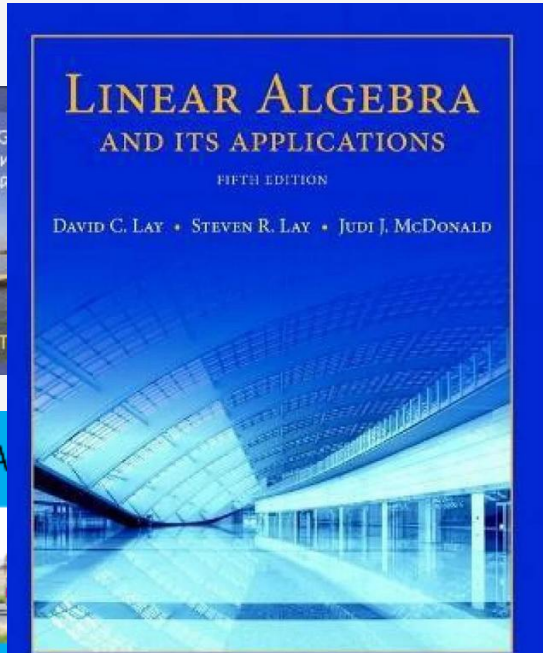
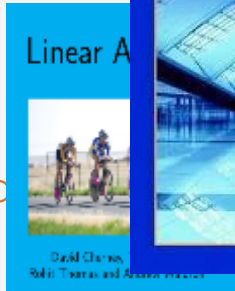
Course Method

- Slides



- Writing on board
- Geometric Interpretation and Intuition
- Notebooks

Course References



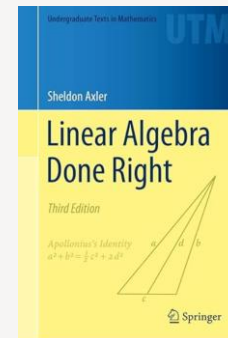
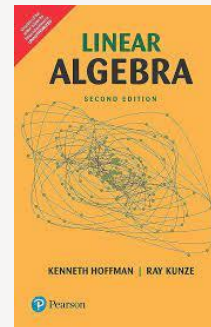
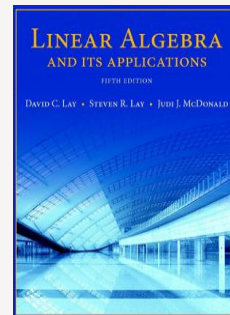
Reach Us!

- Office Room: CE802-CE803
- Email:
 - Hamid R. Rabiee: (rabiee@sharif.edu)
 - Maryam Ramezani: (maryam.ramezani@sharif.edu)
- Course notes, homework and solutions, handouts, and other useful resources are available on the course page:
 - https://quera.org/course/add_to_course/course/23879/ (password: LA1404)
Room: (Sunday & Tuesday: 10:30-12:00)
 - Ebnesina Alef 11
 - <https://vc.sharif.edu/ch/maryam.ramezani> & <https://vc.sharif.edu/ch/rabiee>
- Lead TAs:
 - Asal Meskin (asalmskin82@gmail.com)
- Feedback
 - <https://docs.google.com/forms/d/e/1FAIpQLSdxNVeJwA6lVqrXTeXHqgPz2ne2eIYQJMdNP9fp-wVi-RKNng/viewform?usp=header>

Resources

- Textbooks:
 - Sheldon Axler, Linear Algebra Done Right, fourth edition, 2023
 - Kenneth Hoffman and Ray A. Kunze. Linear Algebra. PHI Learning, 2004.
 - Gilbert Strang. Introduction to Linear Algebra. Wellesley–Cambridge Press, 2016.
 - David C. Lay, Steven R. Lay, and Judi J. McDonald. Linear Algebra and Its Applications. Pearson, 2016.

+Other textbooks and course materials.



Structure of the Course

Lectures

Goal: To introduce concepts in linear algebra, and motivate their use and importance.

Note: We try to cover useful materials in class, but we recommend you reading more!

Exams

Quiz: 6 with dropping lowest

Midterm: 1

Final: 1

Assignments

Purpose: To give you a chance to exercise your mind, and to solidify the concepts introduced to you in class.

Structure: Seven theoretical problems, and Three linear algebra practical problems.

Importance: Not important unless you want to learn the material and get a good grade.

Lecture Notes

- Lecture slide will be uploaded.
- Many times we will write on board, in real-time, during lecture to prove a theory or answer a question or add some additional explanations. It will be your responsibility to take notes.
- Slides links will be provided on site.

Syllabus

1404/07/20	S01	1	Vector Space
1404/07/22	S02		Subspace
1404/07/27	S03		Elementary Row Operations, and Linear Equations
1404/07/29	S04		Row Reduction and Echelon Forms
1404/08/04	S05	2	Linear Independence
1404/08/06	S06		Linear Independence
1404/08/11	S07		Bases & Dimension
1404/08/13	S08		Matrix Rank
1404/08/18	S09		Matrix Rank

Syllabus

1404/08/20	S10	3	Linear Transformation
1404/08/25	S11		Linear Transformation
1404/08/27	S12		Change Basis
1404/09/02	S13		Inverse
1404/09/04	S14		Determinant
1404/09/09	S15		Determinant
1404/09/11	S16	4	Eigenvectors, Eigenvalues, Similarity & Diagonalization
1404/09/16	S17		Eigenvectors, Eigenvalues, Similarity & Diagonalization
1404/09/18	S18		Symmetric Matrices and Quadratic Forms
1404/09/23	S19	5	Matrix Factorization
1404/09/25	Midterm Exam		
1404/09/27	S20		Singular Values and Singular Vectors
1404/09/30	S21		SVD

Syllabus

1404/10/02	S22	6	Inner Product Space
1404/10/07	S23		Inner Product Space
1404/10/09	S24		Inequalities and Orthogonality
1404/10/14	S25		Orthogonality (Gram–Schmidt, etc.)
1404/10/16	S26	7	Norm Space
1404/10/21	S27		Derivatives
1404/10/23	S28		Least squares
1404/11/05	Final Exam		

Assignments

- Theoretical assignments are released at the beginning of each part. You do not need to upload your responses.
- Programming Assignments: 3 points

Assignments

	Release	Deadline
T1	04/07/27	
T2	04/08/06	
T3	04/08/20	
P1	04/08/20	04/09/04
T4	04/09/11	
T5	04/09/23	
P2	04/09/23	04/10/07
T6	04/10/02	
T7	04/10/16	
P3	04/10/16	04/10/30

Exams

Quizzes will be held as the following table on 12:00 for half an hour and will be graded in Grade Scope.

Exam	Time
Quiz 1	1404/08/13 12:00-12:30
Quiz 2	1404/08/20 12:00-12:30
Quiz 3	1404/09/16 12:00-12:30
Quiz 4	1404/09/23 12:00-12:30
Midterm	1404/09/25 10:30-12:45
Quiz 5	1404/10/07 12:00-12:30
Quiz 6	1404/10/21 12:00-12:30
Final	1404/11/05 15:00-18:00

TA Classes

TA classes will be made available to you as recorded sessions through Quera and the course website.



Grading Policy

Quizzes

6 points (6 quizzes each
1.2 points; the lowest
score will be dropped)

Programming Assignments

3 points (each 1 point)

Midterm

6 points (Lectures part 1,2,3,4)

Final

6 points (Lectures part 5,6,7)

Total: 21 Points



Introduction

Lets think about a question!

01

Vector

Basic Concept

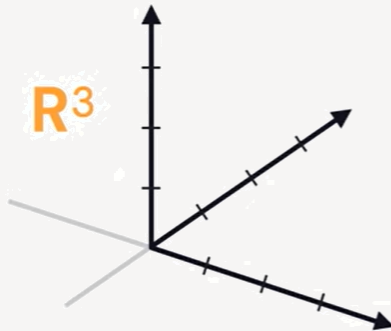


Data Representations (Linear Algebra)

- How can we represent data (images, text, user preferences, etc.) in a way that computers can understand?
- Organize information into a vector!
 - A vector is a 1-dimensional array of numbers.
 - It has both a magnitude (length) and a direction
- The totality of a vectors with n entries is an n-dimensional vector space.

$$V = \begin{matrix} \nearrow \\ = \end{matrix} \begin{bmatrix} -3 \\ 0.7 \\ 2 \end{bmatrix}$$

“3-dimensional space” consists
of all vectors with 3 entries:

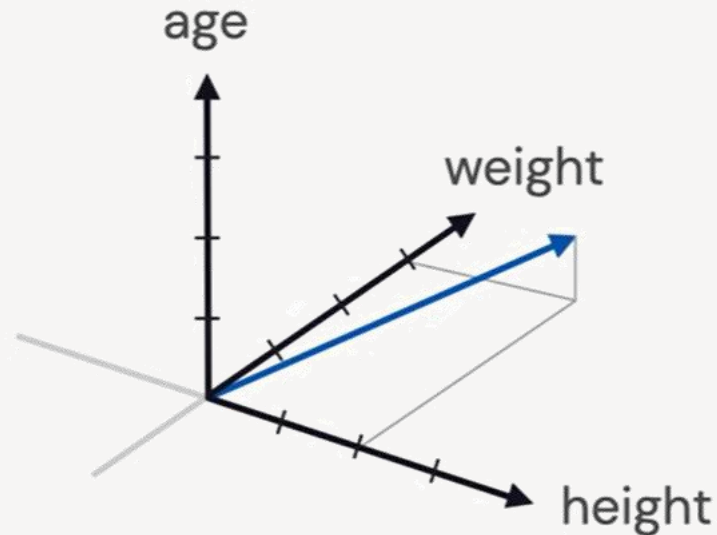


$$\begin{bmatrix} * \\ * \\ * \end{bmatrix}$$

Data Representations (Machine Learning)

- A **feature vector** is a vector whose entries represent the “features” of an object.
- The vector space containing them is called **feature space**.

$$P = \begin{bmatrix} 64 \\ 131 \\ 24 \end{bmatrix} \begin{array}{l} \text{height} \\ \text{weight} \\ \text{age} \end{array}$$





02

Equation with Matrix and Vector Format



House Pricing Example



Price Problem



\$ 70'000



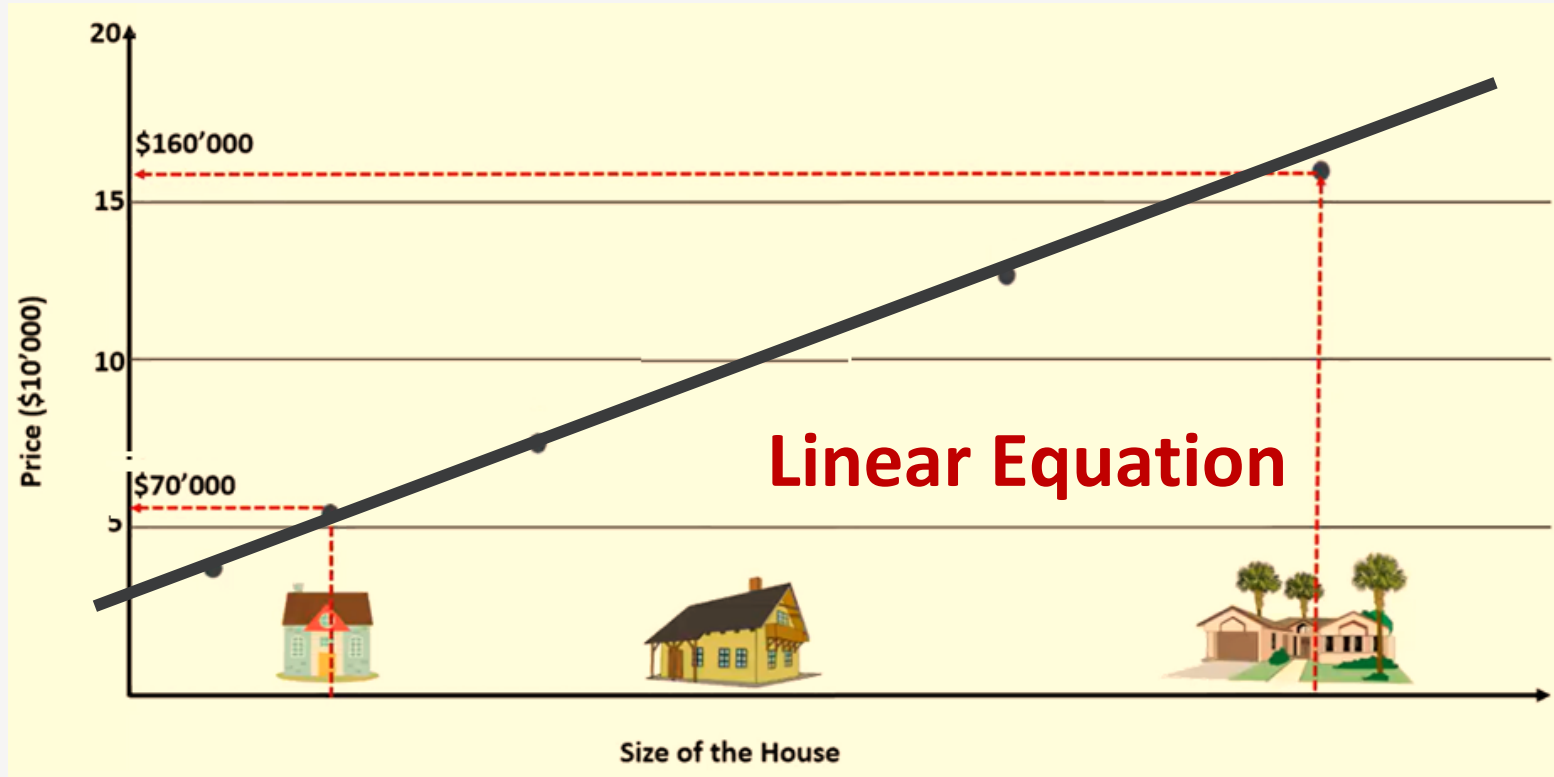
?



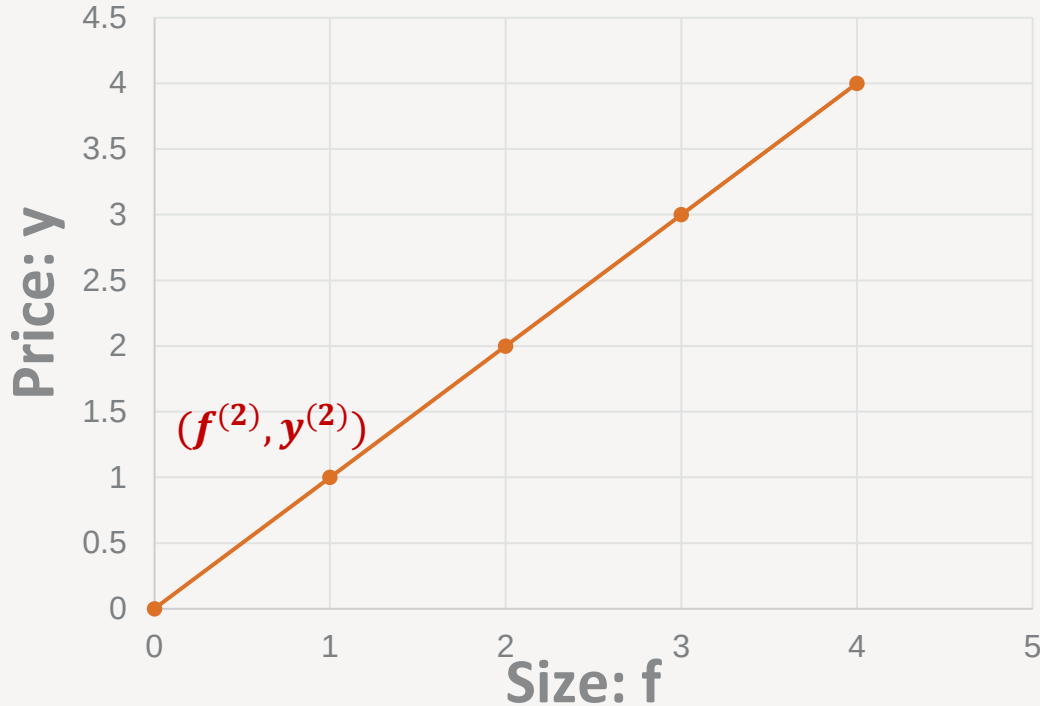
\$ 160'000



Price Problem



Linear Equation without offset



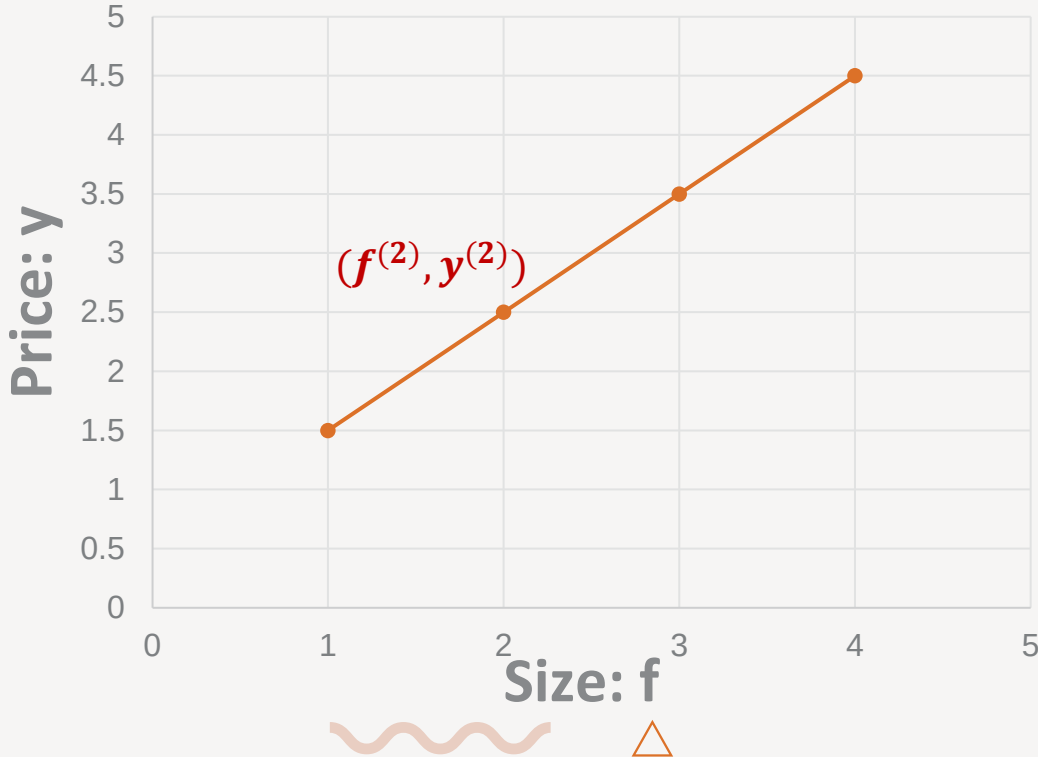
$$y = af$$

How to convert it to matrix-vector multiplication?

$$Ax = b$$



Linear Equation with offset



$$y = af + c$$

How to convert it to matrix-vector multiplication?

$$Ax = b$$

House Features

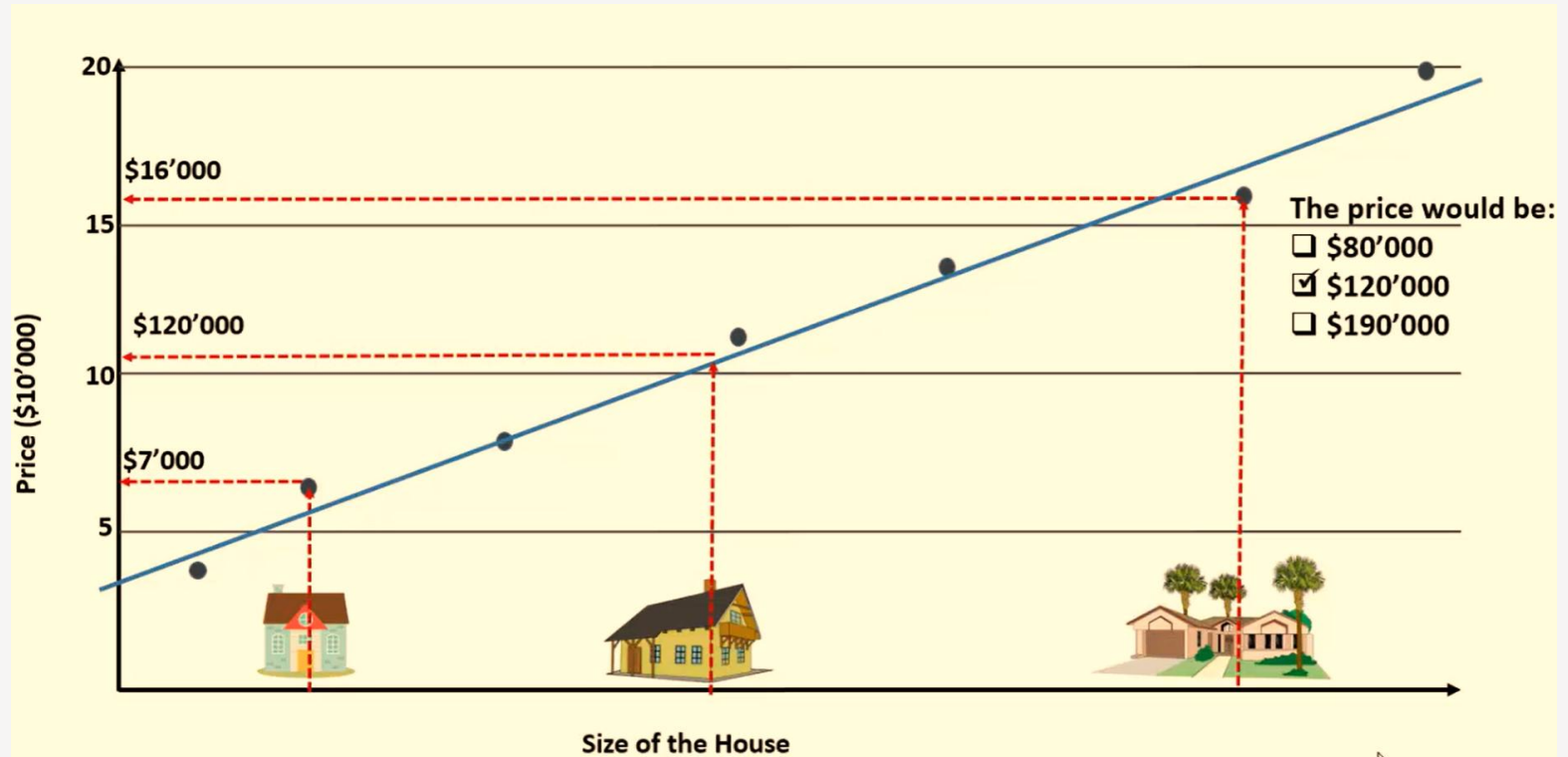
- #Room
- Size
- #Bedroom
- Age
- Address features: Street, Alley, ...
- Size of part1, part2, part3, part4
- Floors
- #Bathrooms

N number of training data with M features:

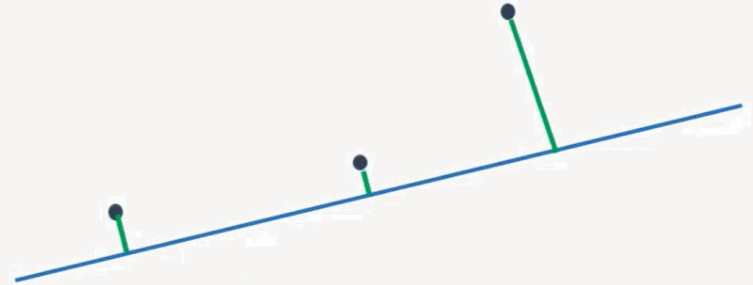
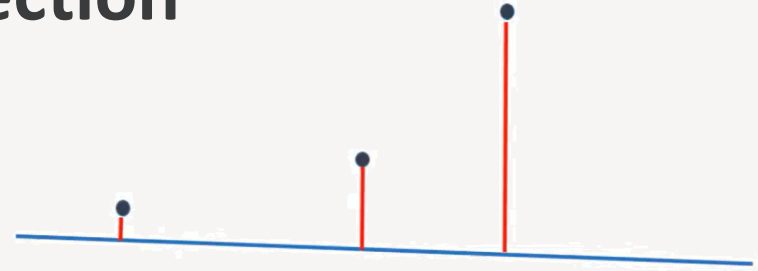
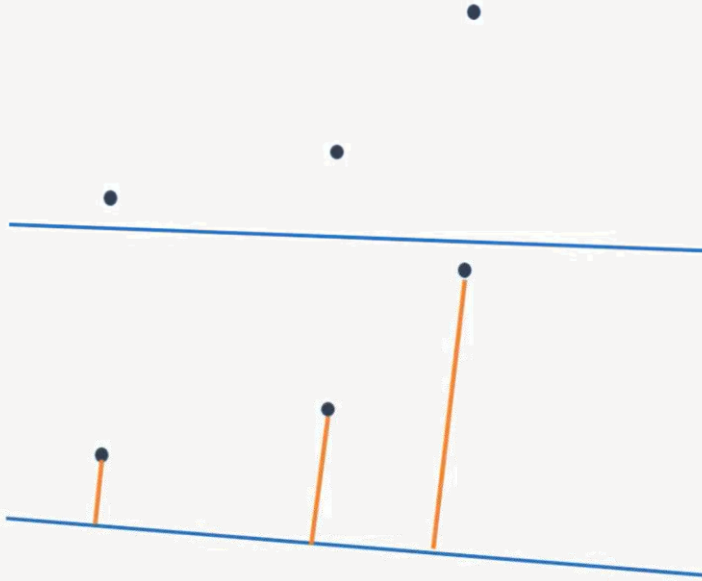
$$A_{N \times M} x_{M \times 1} = b_{N \times 1}$$

Linear Equation

Least Squares Error Correction



Least Squares Error Correction



Error 1: 

Error 2: 

Error 3: 

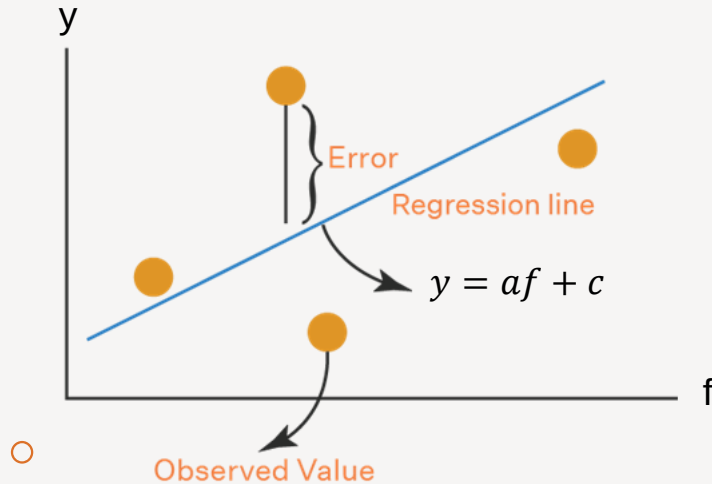
Least Squares

Least Square Method

- Objective for “m” features:

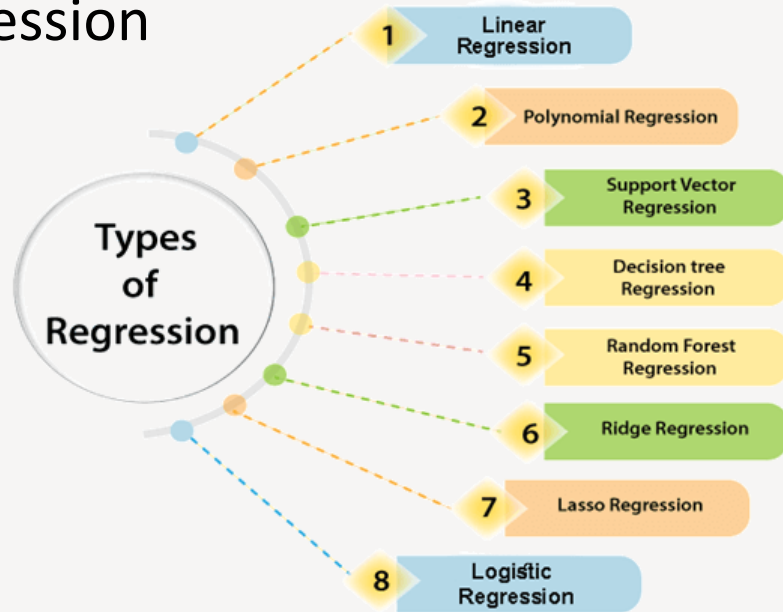
$$\hat{y} = a_1 f_1 + a_2 f_2 + \cdots + a_m f_m + c$$

$$\min ||y - \hat{y}||$$



Linear Algebra and Machine Learning Application

- $Ax = b \rightarrow x = A^{-1}b$ Inverse of Matrix/Pseudo Inverse
- Regression



02

Road Map

Step by Step



